

tf.batch_to_space Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/batch_to_space

BatchToSpace for N-D tensors of type T.

Function Signatures

```
tf.batch_to_space( input, block_shape, crops, name=None )
```

Code Examples

```
tf.batch_to_space(  
input, block_shape, crops, name=None  
)
```

```
tf.batch_to_space(  
input, block_shape, crops, name=None  
)
```

```
[[[1]]],  
[[[2]]],  
[[[3]]],  
[[[4]]]]
```

```
[[[1]],  
 [[2]],  
 [[3]],  
 [[4]]]
```

```
x = [[[1], [2]],  
      [[3], [4]]]
```

```
x = [[[1], [2]],  
      [[3], [4]]]
```

```
[[[1, 2, 3],  
  [4, 5, 6],  
  [7, 8, 9],  
  [10, 11, 12]]]
```

```
[[[1, 2, 3],  
  [4, 5, 6],  
  [7, 8, 9],  
  [10, 11, 12]]]
```

Documentation

BatchToSpace for N-D tensors of type T.

This operation reshapes the "batch" dimension 0 into $M + 1$ dimensions of shape `block_shape + [batch]`, interleaves these blocks back into the grid defined by the spatial dimensions `[1, ..., M]`, to obtain a result with the same rank as the input. The spatial dimensions of this intermediate result are then optionally cropped according to `crops` to produce the output. This is the reverse of `SpaceToBatch` (see `tf.space_to_batch`).

- Reshape input to reshaped of shape: [block_shape[0], ...,

block_shape[M-1], batch / prod(block_shape), input_shape[1], ...,
input_shape[N-1]]

- Permute dimensions of reshaped to produce permuted of shape

[batch / prod(block_shape), input_shape[1], block_shape[0], ...,
input_shape[M], block_shape[M-1], input_shape[M+1], ...,
..., input_shape[N-1]]

- Reshape permuted to produce reshaped_permuted of shape

[batch / prod(block_shape), input_shape[1] * block_shape[0], ...,
input_shape[M] * block_shape[M-1], input_shape[M+1], ...,
input_shape[N-1]]

- Crop the start and end of dimensions [1, ..., M] of

reshaped_permuted according to crops to produce the output
of shape:

[batch / prod(block_shape), input_shape[1] *
block_shape[0] - crops[0,0] - crops[0,1], ..., input_shape[M] *
block_shape[M-1] - crops[M-1,0] - crops[M-1,1], input_shape[M+1],
..., input_shape[N-1]]

Examples:

- For the following input of shape [4, 1, 1, 1],
block_shape = [2, 2], and crops = [[0, 0], [0, 0]]:

[[[1]]],

[[[2]]],

```
[[[3]]],  
[[[4]]]]
```

The output tensor has shape [1, 2, 2, 1] and value:

```
x = [[[1], [2]],  
[[3], [4]]]]
```

- For the following input of shape [4, 1, 1, 3],

block_shape = [2, 2], and crops = [[0, 0], [0, 0]]:

```
[[[1, 2, 3]],  
[[4, 5, 6]],  
[[7, 8, 9]],  
[[10, 11, 12]]]
```

The output tensor has shape [1, 2, 2, 3] and value:

```
x = [[[1, 2, 3], [4, 5, 6 ]],  
[[7, 8, 9], [10, 11, 12]]]]
```

- For the following

input of shape [4, 2, 2, 1],

block_shape = [2, 2], and crops = [[0, 0], [0, 0]]:

```
x = [[[1], [3]], [[ 9], [11]]],  
[[[2], [4]], [[10], [12]]],  
[[[5], [7]], [[13], [15]]],  
[[[6], [8]], [[14], [16]]]]
```

The output tensor has shape [1, 4, 4, 1] and value:

```
x = [[[1], [2], [ 3], [ 4]],  
[[5], [6], [ 7], [ 8]],  
[[9], [10], [11], [12]],
```

[[13], [14], [15], [16]]]

- For the following input of shape

[8, 1, 3, 1],

block_shape = [2, 2], and crops = [[0, 0], [2, 0]]:

x = [[[[0], [1], [3]]],

[[[0], [9], [11]]],

[[[0], [2], [4]]],

[[[0], [10], [12]]],

[[[0], [5], [7]]],

[[[0], [13], [15]]],

[[[0], [6], [8]]],

[[[0], [14], [16]]]]

The output tensor has shape [2, 2, 4, 1] and value:

x = [[[[1], [2], [3], [4]],

[[5], [6], [7], [8]]],

[[[9], [10], [11], [12]],

[[13], [14], [15], [16]]]]

For the following input of shape [4, 1, 1, 1],

block_shape = [2, 2], and crops = [[0, 0], [0, 0]]:

The output tensor has shape [1, 2, 2, 1] and value:

For the following input of shape [4, 1, 1, 3],

block_shape = [2, 2], and crops = [[0, 0], [0, 0]]:

The output tensor has shape [1, 2, 2, 3] and value:

For the following

input of shape [4, 2, 2, 1],

block_shape = [2, 2], and crops = [[0, 0], [0, 0]]:

The output tensor has shape [1, 4, 4, 1] and value:

For the following input of shape

[8, 1, 3, 1],

block_shape = [2, 2], and crops = [[0, 0], [2, 0]]:

The output tensor has shape [2, 2, 4, 1] and value:

Returns